

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019**Privacy Preservation and Signature Aggregation for Medical Data**

Ketan Atri, Yash Mahesh Bangera*, Vishwas S. Deshmukh, Dr. Vaidehi Vijayakumar

*Vellore Institute of Technology, Chennai ; *Mother Teresa Women's University, Kodaikanal*

Abstract

Remote healthcare monitoring systems have been successful and are present at the vanguard of the technological revolution of the 21st Century. These systems make use of wireless body area network (WBAN) as one of their major components that is responsible for recording the medical parameters of a patient's health and transmit them to the server where they can be diagnosed. These networks, sometimes also make use of local processing units (LPUs) such as your mobile devices for transmission of the recorded data. Unfortunately, this wireless transmission of data increases the number of risks involved with the sensitive recorded data. This can lead to the violation of user's privacy rights and also causes a decline in the system's integrity and authenticity. This paper presents a novel security and signature aggregation scheme implemented using the Arduino Uno microcontroller, the MAX30100 SPO2 and Pulse Oximetry sensor, and the AD8232 sensor. The Privacy Preservation and Signature Aggregation Scheme (PPSA) for Medical Data that we present in this paper helps provide security against cyber attacks, thus strengthening the integrity and authenticity of the medical data. The Privacy Preservation part of the overall scheme has been implemented using the Blowfish and the Elgamal Encryption schemes, that helps the system to maintain the security of the system, whereas the Signature aggregation scheme involves the use of a co-Decisional Group Diffie-Hellman Signature scheme that helps the system to verify the authenticity of the medical data that is being transmitted within the system. The PPSA scheme, when implemented, was found to give correct results and provide the system with reliable security.

© 2019 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019.

Keywords: WBAN; LPU; CPS; RHMS; PPSA; Elgamal Encryption; Blowfish Encryption; Signature Aggregation

1. Introduction

Cyber-physical systems (CPS), wireless sensors and other communication technologies have been seamlessly integrated into our modern-day lives that have in turn led to the development of many applications in areas that require constant monitoring of the environment. The Wireless Body Area Network (WBAN) refers to one such system that makes use of tiny sensors that can be attached to the patient's body and are responsible for measuring specific physiological parameters such as SPO2 percentage, pulse, temperature, electrocardiogram values and blood pressure of the body. They also send this recorded medical data to the healthcare professional for further diagnosis. However, it is of much regret that these sensors, due to their association with the human body, are deployed in hostile and mobile

* Yash Mahesh Bangera et al. ,Tel.: +91 9892279995, email: yashmahesh.bangera2015@vit.ac.in

environments that not only increases the communication overhead but also leads to the insecurity of the medical data that is being transmitted wirelessly that remains susceptible to a long line of attackers such as data snoopers, data forgers as well as data misusers.

To steer clear of such malicious adversaries, the Health Insurance Portability and Accountability Act (HIPPA) has provided health organizations with regulations for data privacy and for safeguarding the medical data. This act has emerged into the vanguard with the proliferation of health data breaches caused by cyber-attacks on medical healthcare systems and has, therefore, forced healthcare organizations to view their remote healthcare monitoring systems (RHMS) under a legal standpoint. In the absence of such measures, cases and complaints can be lodged against those organizations that are found to be in breach. These cases prove to be very costly and fatal for such healthcare establishments.

Our project aims towards eliminating threats that such remote medical systems can face during the transmission or recording of the patient's data. For implementing such security measures, we make use of data encryption. This is implemented using the two-layer Privacy Preservation scheme that includes the famous Blowfish Encryption paired along with the Elgamal Encryption scheme. The incorporation of this scheme ensures data confidentiality and increases resilience of the system towards data eavesdroppers and data snoopers. There also rises another field of constraints due to the use of sensors in such remote medical systems. Sensors that are used in the WBAN have low-computational power and a low battery life. Fortunately, our proposed scheme provides solutions to these problems too by making use of the Arduino microcontroller and local processing units (LPUs) that have on-board processing capabilities which help in reducing overall computational costs as well as the communication overhead. The last part of the project deals with the incorporation of a data validation and a data verification protocol for the medical system. This is brought about by the Signature Aggregation scheme that is performed at the LPU. This creates space for reduction in the number of data hackers that manipulate the recorded medical data that could lead to incorrect diagnosis and even have severe consequences. Altogether, the scheme can therefore be termed as the Privacy Preservation and Signature Aggregation scheme (PPSA).

The rest of the paper is organized as follows:

- Section 2 describes the Literature survey. Section 3 lays out the details of the proposed system. Section 4 gives the experimentation details about our PPSA scheme. Section 5 discusses the implementation and results. Section 6 contains performance evaluation details. Section 7 concludes the paper and also discusses the future direction.

2. Literature Survey

2.1. Wireless Body Area Network (WBAN)

A wireless body area network sometimes also referred to as the body area network (BAN) or the medical body area network (MBAN) is a wireless network of sensing devices with on-board processing and computing capabilities. Challenges of the WBAN are mentioned below:

•Data Quality •Data Integrity •Low Hardware Capacities •Data Confidentiality

2.2. Computational Diffie Hellman Assumption (CDH)

The computational Diffie-Hellman assumption is the reason as to why we can select the Elgamal Encryption scheme in our novel scheme. It allows us to share the public key of the Elgamal Encryption scheme without having to worry about the private key being computed from that. This is because the assumption states that for a cyclic group with a generator 'g' and randomly chosen values 'a', 'b' where 'a' and 'b' belong to the range $\{0, q-1\}$, where q is the order of the group; if we are given the values (g, g^a, g^b) then it is computationally intractable to compute the value g^{ab} . We are making use of the generator g, the group F, the order q and $h = g^a$, where 'a' is the private key. If the attacker can determine 'a' from the encrypted message, then he can determine the private key, but we make use of g^{ak} as the encrypted message which makes it impossible for decryption using the current resources.

2.3. Decisional Diffie Hellman Assumption (DDH)

Consider that we have a multiplicative cyclic group G of order q with a generator g. As mentioned in the above sub-section, if we choose $a, b \in Z_q$, with given values of g^a and g^b , the value g^{ab} looks like a random element in the group G. This notion is proven by saying that the following two probability distributions are indistinguishable when computed:

- (g^a, g^b, g^{ab}) , where a,b are randomly and independently chosen from Z_q .
- (g^a, g^b, g^c) , where a,b,c are randomly and independently chosen from Z_q .

2.4. Decisional Co-Diffie-Hellman problem (co-DDH)

Let's say that we have two multiplicative cyclic groups G_1 and G_2 that have h and g_2 as their generators. The decisional co-Diffie Hellman problem says that given the values g_2 and $(g_2)^a$ that belong to the group G_2 and h, h^b that belongs to the group G_1 , we give the output yes if a is equal to b and no otherwise. When the answer is yes, we can say that the tuple $(g_2, (g_2)^a, h, h^b)$ is a co-Diffie-Hellman-Tuple. The properties of this tuple are used in the signature aggregation scheme of our project.

2.5. Bilinear Maps

Let's say that we have two groups G_1 and G_2 such that the target group G_T follows the following property: $|G_1|=|G_2|=|G_T|$. A bilinear map, in this case, is defined as a map e that is denoted as $e: G_1 \times G_2 \rightarrow G_T$. More details on bilinear maps are given in [1] and [2].

2.6. Data Collection Rate

The AD8232 ECG readings are sent after each millisecond, whereas the Pulse and SPO2 readings from the MAX30100 are sent every second. The data is saved at the Bluetooth buffer of the mobile device and is read at regular intervals of 1 second each.

2.7. Privacy Preserving and Signature Aggregation (PPSA)

Unlike the common forms of aggregation in [3] and [4] which aggregate the data directly, we use a scheme that aggregates the signatures instead of the data itself. This process involves signing of the data using a hash function and a secret key and then aggregates the signature before sending it to the medical server. At the medical server, this signature is verified, thus incorporating the data verification functionality of the system.

2.8. Ring Based Privacy Preservation and Aggregation scheme (RiPPAS)

RiPPAS is a ring-based privacy preserving aggregation scheme that makes use of homomorphic encryption techniques for encryption of the data recorded from wireless sensors. More details about the scheme are presented in [5].

2.9. Secure Privacy Preserving Data Aggregation Scheme (SPPDA)

It is a privacy preserving data aggregation scheme that uses the bilinear Elgamal cryptosystem as its main component. More algorithm details are presented in [1].

3. Proposed System

3.1. Circuit and Hardware components

Our PPSA scheme has been tested on a prototype of a remote healthcare monitoring system in which an Arduino microcontroller, a MAX30100 sensor, an AD8232 ECG sensor and a Bluetooth module as the components of the Body Area Network. The Arduino Uno microcontroller provides onboard processing, ease of use and cost reduction abilities to our system. The MAX30100 sensor is an integrated chip that is responsible for detecting the pulse and SPO2 percentage of a patient. It also helps in detecting heart rate signals and provides optimized power consumption. Lastly, the AD8232 sensor is used for measuring the electrical activity of the heart. This can be charted onto an ECG or electrocardiogram, which can be further used for diagnosing heart conditions. This is also called AD8232 Heart Rate Monitor.

3.2. Interface Schematics

The sensors record the medical data and send it to the Arduino, where it is encrypted upon reception. The signing of the data also occurs at this microcontroller. This data is then sent to the LPU via a Bluetooth module that is connected to the Arduino. The ECG readings are sent after each millisecond, whereas the Pulse and SPO2 readings are sent every second. The data is saved at the Bluetooth buffer of the mobile device and is read at regular intervals of 1 second each. The LPU pushes the data to Firebase realtime database, which is a cloud-hosted No-SQL Database. The values are stored there in an encrypted format along with their signatures. The server side uses PHP programming to read the data from the Firebase database, perform verification and decryption of data before displaying it to the end-user. The integrity of the data is verified and the signatures are aggregated at the LPU before sending the data over to the server. Only authorized users are allowed access to the confidential data present on the server. Even among the authorized users, there is clearance level which is determined by the server using a login ID. Doctor's login IDs are given keys to all of the patient's data, whereas those of other personnel only have keys to a patient's sensor readings. Personnel other than a doctor cannot view a patient's private information such as name, age and the gender. Upon entering the login ID and password, the server asks for the patient ID whose data the user wishes to see. It then decrypts the data of that patient for displaying it on the user's screen. Note that decryption is performed in real-time and as such, there is no storage of decrypted data, hence the patient's readings remain safe at all times.

4. Experiment

The components of the case study that has been undertaken in this paper have been discussed in the further sections.

4.1 Architecture

The privacy preservation and signature aggregation scheme is supported by a three part system that consists of the following:

- Body Area Network (BAN) •Personal Area Network (PAN) •Medical Server.

The overview of the system is described in figure 1 given below:

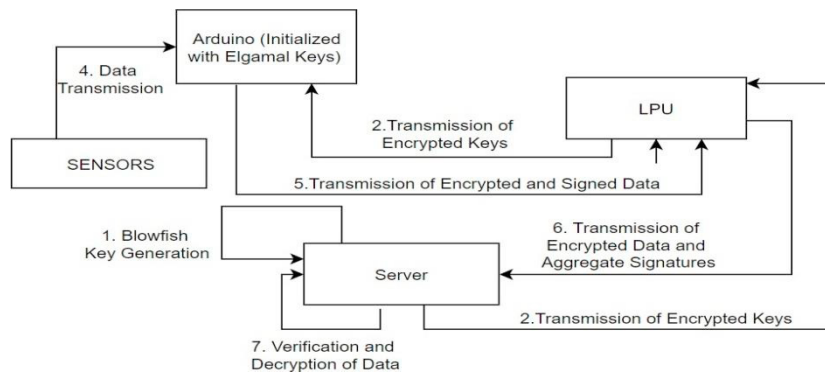


Fig. 1. Architecture of the system

These system components are described in the next sub-sections.

4.1.1. Body Area Network

The Body Area Network (BAN) comprises of the medical sensors that are placed on the body of the patient which is responsible for recording and transmission of medical data from the patient's body to the local processing unit (LPU), which in this case can be a mobile device or a personal digital assistant (PDA). We have made use of the following sensors in the implementation of our project: MAX30100 Pulse and SPO2 percentage sensor AD8232 sensor

The fact that these sensors are deployed in environments that have a paucity in the storage, computational and other resources such as battery life. In a scenario where these sensors are used as part of the WBAN where wireless transmission is required, the chances of cyber-attacks increase highly and might lead to an incorrect diagnosis of the patient's data that might prove to be fatal for some patients. Moreover, there also exists a legal necessity which makes it mandatory to ensure the privacy of the data. Hence, the data that is recorded using these sensors must be encrypted and signed before it could be sent to the LPU. However, there is also the fact that these sensors have a low battery life and storage power. Hence, encryption cannot be performed by using the sensors' on-board processing capabilities. Thus, we need an Arduino microcontroller connected to these sensors that performs the encryption and the LPU that applies signatures on this encrypted data which is then followed by the process of transmission, which are described further.

4.1.2. Personal Area Network

The personal area network consists of the Local Processing Unit which can either be a personal digital assistant (PDA) or a mobile device carried by the user. In our case study, we use the mobile device carried by the user as the Local Processing Unit (LPU). The local processing unit possesses more resources as compared to the medical sensors that are being used in the project. The data that is being handled at this unit is already encrypted at the Arduino in the previous stage of this project. The mobile device carried by the user is connected by the means of Bluetooth technology to the Arduino microcontroller. This device is responsible for collecting the encrypted data from the Arduino and creating aggregate signatures on the data that are to be sent to the server. As we can see the aggregation employed here is computationally intensive, however unlike [6] which employs deep learning, it does not require too much of resources. Even [8] is not suitable for our cause as it relies on Evolutionary algorithm which again is computationally intensive. So the choice of mobile device fits perfectly for our usage. These signatures ensure the resilience of the system against data forgery attacks and make the system more authentic. The signature aggregation also helps in reducing communication costs of the system.

4.1.3. Server

The medical server is very powerful in terms of energy, computational power, and storage. The medical server also allows access to the practitioners, doctors, researchers, or nurses that have the proper credentials or authority to get access to the data of the patient in context. The local processing unit, as seen earlier, is responsible for creating the aggregate signatures, after which the data is communicated wirelessly to the medical server. The medical server performs two major functions. First, the last part of the signature aggregation scheme that is the aggregate verification of the data is performed at this server. The second task performed at the server side is the decryption of the data that was encrypted at the Arduino microcontroller. After this, the data is ready to be used through this server by only the authorized individuals.

Following are the major details of the overall system design that we have undertaken in our case study:

- **Key generation and sharing:** The first phase starts with the remote medical server bootstrapping the whole system. There exists a key generation function stored at the server end which is utilized by the server to generate the public and private key pairs that are used in the Elgamal and Blowfish Encryption at the Arduino microcontroller. The key generation is performed at the medical server because it has a better performance in terms of battery, storage and computational power than the sensors, the local processing unit or the Arduino microcontroller for that matter.
- **Data generation, encryption and signing:** The health data is generated by the sensors and is encrypted by the Arduino microcontroller and then it is sent to the local processing unit (LPU). The health data is reported to the local processing unit at regular epochs of time. For example, the ECG data is transmitted at every millisecond whereas the Pulse Oximetry sensor data is transmitted at every second. However, before the transmission of the data it is encrypted using the Blowfish algorithm by the Arduino Microcontroller, it is also signed by the same.
- **Aggregation of signatures:** The Arduino performs the encryption and signing of the data transmitted health data. The LPU, upon reception of the messages first verifies the signatures, and then it aggregates the signatures before transmitting it to the server. No decryption is performed by the LPU which saves both processing power and time.
- **Decryption and Verification:** After generating the aggregate signature σ , the local processing unit sends it to the medical server where it is first verified and then the final part of the decryption of the message is performed. This decrypted data, however can only be accessed by the proper personnel according to their authorization and clearance level. For example, Doctor A can access all data of the patients including the name, age, location and other personal information. But, a researcher cannot access the same data. He/she is only allowed to view the numerical medical records without the personal information. This also might require the proper permission from the hospital's authority.

4.2 Algorithm

As explained in the earlier sub-section, we have a three-phase system that supports our aggregation scheme. Single layered alternatives such as [7] are also a viable option; however, our focus is on real-time encryption and transmission. In our case study, the data is first collected at the wireless sensors which is sent to the Arduino microcontroller for our newly-proposed two-level encryption scheme. This is then followed by the Local processing unit receiving this encrypted data at regular epochs which then aggregates it to enable a reduction in communication costs of the system. Lastly, the medical server performs the verification and decryption of this signed data. For our entire system, we require three algorithms that need to be implemented which are as follows:

- Blowfish Encryption Scheme • Elgamal Encryption Scheme • Aggregate Signature Scheme

4.2.1. Blowfish Encryption Scheme

Blowfish encryption scheme provides a good rate at which the data is encrypted and there is no effective analysis of the cryptosystem till date. It is a symmetric key-based encryption scheme. This encryption scheme comprises of three stages: •Encryption •Decryption •Key Generation

4.2.2. Elgamal Encryption Scheme

The keys that are generated by the Blowfish Encryption Scheme are used as input to the Elgamal Encryption Scheme for adding a second layer of security that leads to the rise of a novel scheme to the healthcare monitoring system. Elgamal is an asymmetric key-based encryption scheme which is why it is used to provide the additional layer of security by using it to encrypt the keys of Blowfish algorithm before they are transmitted to the Arduino microcontroller. The entire Elgamal Encryption scheme can be seen as the following parts:

- Key Generation •Encryption •Decryption. These parts are discussed further in detail in [1].

4.2.3. Aggregate Signature Scheme

This signature scheme is based upon the work done in [1]. The signature aggregation is performed on the local processing unit where the encrypted data is first received. The aggregate signature scheme allows for the creation of signatures on arbitrary distinct messages $M_i \in \{0, 1\}^*$. An individual signature σ_i is an element of cyclic group G_1 . The base groups G_1 and G_2 , their respective generators g_1 and g_2 , the computable isomorphism from G_2 to G_1 , and the bilinear map $e: G_1 \times G_2 \rightarrow G_T$, with target group G_T , are the system parameters. The aggregate signature scheme comprises of the following algorithms: •KeyGen •Sign •Verify •Aggregate •AggregateVerify

5. Implementation and Results

Our case study consists of a prototype of a remote healthcare monitoring system that comprises of two sensors - the MAX30100 sensor and AD8232 ECG sensor that are connected to the Arduino Uno microcontroller. These two sensors record the pulse and SPO2 percentage, and the ECG and heart rate values of the patient respectively. So, in the end, we have a dataset that includes the pulse, SPO2 percentage and the ECG values of the patient that is to be diagnosed. We make use of these parameters to test our Privacy Preservation and Signature Aggregation (PPSA) Scheme. First, these parameters are recorded at the Arduino microcontroller. The Blowfish Encryption scheme is responsible for key generation. These keys are then encrypted using the Elgamal Encryption scheme. We now have keys that will be given as input into the 16-round cipher of the Blowfish Encryption scheme. This takes input of a plain-text, i.e., the recorded medical data and will give ciphertext as the output. This encrypted data is now sent to the local processing unit (LPU) where it is signed to make sure that it becomes resilient to data forgery attacks. The signature aggregation scheme is applied at the LPU, which is a mobile device connected via the Bluetooth module to the Arduino board, in our case study. The encrypted data is now signed and this aggregated signed data is sent to the remote medical server where it is verified. Upon verification, the decryption of the data is carried out, which is displayed only to the authorized personnel through this server.

We were successful in collecting the data in the first phase and encrypting it using our two-level encryption scheme. Then we sent it to the Bluetooth connected mobile device where it was signed and an aggregated signature was created. This was then sent to the remote medical server where it was then verified and then decrypted successfully getting the original data that was recorded. This data could be viewed only by the proper personnel whose credentials are recorded in the database. We consider the following scenarios and evaluate the performance of our system under the following conditions:

5.1 Condition1: When the Arduino has to be connected via Bluetooth to the local processing unit.

The Arduino can be misused by other local processing units present in the region of connectivity via the Bluetooth connection. The advantage we have in this situation is that our recorded data reaches the mobile device or LPU in an encrypted format. So, data snooping is unsuccessful on our system.

5.2 Condition2: Men in the middle can inject false data into the transmitted batches of data due to wireless connectivity and hostile environments.

The signature aggregation scheme allows for robustness against the data forgery attacks by verifying the sign of the received data at the medical server.

5.3 Condition3: Cryptanalysis of the cipher-text can be performed.

Blowfish encryption scheme is a big advantage to our system because the keys that it generates are so complex that they appear random even after effective cryptanalysis. So, the overall security level is tough for anybody to attack the system very easily.

The data is sent at regular epochs to the local processing unit or the mobile device where it is signed and the signature is aggregated. Each epoch sends an 8-bit key. So, in 2 epochs we will send 16-bits of data. We now perform signature aggregation on this 16-bit data to get 8-bit of data that is to be sent. In this process, we end up saving 8 bits of data. In 1 milli second, 1000 records are saved in the system that need to be signed by the scheme. If we calculate the amount of data that is stored in the overall process, we can be sure that a lot of data is being saved on a per transaction basis.

6. Performance Evaluation

The project performs better when compared to two other schemes that already deal with the situation of privacy preservation and aggregation of medical data. The RiPPAS as discussed in [5] performs hop-to-hop encryption, whereas our PPSA scheme follows end-to-end encryption that gives 70 % better performance capabilities as mentioned in [1]. The SPPDA scheme as discussed in [1] is not possible to perform at the Arduino since the Bilinear Elgamal Cryptosystem requires a lot of computational and processing power to encrypt the medical data recorded by the WBAN.

7. Conclusion and Future Work

The project aimed towards resolving two major problems in the field of remote healthcare monitoring systems. The first problem is the insecurity of the communication and the second one is the low levels of battery life of the sensors used for wireless transmission. The first aspect of the problem is resolved by using a two-layered encryption scheme, whereas the second aspect of the problem is resolved using a signature aggregation scheme. We have been successful not only in strengthening the security of the system, but also achieved an increase in the rate of computation during encryption of the medical data being recorded. The Blowfish Encryption scheme that we have implemented in the system does not hamper with the speed of the overall scheme and also increases the overall security factor of the recorded data. This makes sure that this novel scheme can be used for real-time applications also. However, there is a downside to all of this which makes the Blowfish scheme more susceptible to breaches due to its symmetric nature. To avoid this crisis, we have made use of the ElGamal Encryption scheme instead for key distribution of the Blowfish Encryption Scheme. ElGamal, being an asymmetric scheme is relatively slower during encryption and decryption. Thus, by making a compromise on both levels of the scheme and implementing both Blowfish and Encryption scheme, we ensure an increase in both the speed as well as the security. To make the entire system even more robust, we have also incorporated a signature scheme, which prevents data forgery attacks. As far as the aggregation of the data is concerned, the main challenge was to reduce the size of the data without losing sensitive information. The Aggregate Signature Scheme is implemented on the actual data, where we aggregate the signature which is applied onto the data for maintaining the authenticity of data. Since original data remains intact during this aggregation scheme, we are able to preserve all the sensitive details of the recorded data as well as reduce the amount of data which needs to be transmitted. As part of future possible work, an additional layer of hardware i.e. Arduino Uno board can be removed from the system if we can incorporate tiny processors with the sensors. These can be initialized at the factory to have the public and private keys hardcoded into them, and then only need to perform encryption of data before its transmission. Another possibility of improvement exists in the aggregation scheme. If the sensors can detect when the data to be transmitted is critical enough for a real-time transmission, then they can limit the data transmission at other times by sending data periodically. We have implemented this scheme only for the temperature sensor as it is easy to determine critical conditions in that case; however, a future improvement can be employing the same idea on other sensors such as the ECG sensors which requires more in-depth analysis to determine whether or not it is in a critical state.

References

- [1] ANEES ARA, Jr., (Member, IEEE), MZNAH AL-RODHAAN, YUAN TIAN, AND ABDULLAH AL-DHELAAN “A Secure Privacy-Preserving Data Aggregation Scheme Based on Bilinear ElGamal Cryptosystem for Remote Health Monitoring Systems” *IEEE access* 32 pp. 12601-12617, 2017.
- [2] D.Boneh, C.Gentry, B.Lynn, and H.Shacham “Aggregate and verifiable encrypted signatures from bilinear maps” *Advances in Cryptology, EUROCRYPT*. Berlin, Germany: Springer, 2003
- [3] Misra, Sudip and Chatterjee, Subarna. “Social choice considerations in cloud-assisted WBAN architecture for post-disaster healthcare: Data aggregation and channelization.” *Information Sciences* 284. 95-117. 10.1016/j.ins.2014.05.010, 2014
- [4] Mohamed Ben Haj Frej, Khaled Elleithy “Secure Data Aggregation Model (SDAM) in Wireless Sensor Networks” *2015 IEEE 14th International Conference on Machine Learning and Applications (ICMLA)*, Miami, FL , 2015, pp. 330-334
- [5] Kejia Zhang , Qilong Han , Zhipeng Cai and Guisheng Yin “RiPPAS: A Ring-Based Privacy-Preserving Aggregation Scheme in Wireless Sensor Networks” *Sensors (Basel)* 17(2), 2017 Feb 7
- [6] Nasim Heydari, Behrouz Minaei –Bidgoli “Reduce energy consumption and send secure data wireless multimedia sensor networks using a combination of techniques for multi-layer watermark and deep learning” *IJCSNS, International Journal of Computer Science and Network Security*, February 2017
- [7] Yang, G., Sen, L., Xu, X., Dai, H., & Yang, Z. “Precision-Enhanced and Encryption-Mixed Privacy-Preserving Data Aggregation in Wireless Sensor Networks” *IIJDSN*, 9, 2013
- [8] Ms. Sheetal , V. Chate, Prof. S. B. Javheri “Implementation of Clustering For Data Aggregation Using Evolutionary Algorithm” *2006 IJEDR*